11. Map Coloring (Constraint Satisfaction Problem)

```
def is_safe(node, color, assignment, graph):
    # Ensure neighbors do not share the same color assignment
    for neighbor in graph.get(node, []):
        if neighbor in assignment and assignment[neighbor] == color:
            return False
    return True

def map_coloring_csp(graph, colors, assignment, nodes, index=0):
    if index == len(nodes):
        return True

    current_node = nodes[index]
    for color in colors:
        if is_safe(current_node, color, assignment, graph):
            assignment[current_node] = color

            if map_coloring_csp(graph, colors, assignment, nodes, index + 1):
                return True

            # Backtrack step
            del assignment[current_node]

    return False

if __name__ == "__main__":
    # Map layout represented as an adjacency network
    map_graph = {
        'WA': ['NT', 'SA'],
        'NT': ['WA', 'SA', 'Q'],
        'SA': ['WA', 'NT', 'Q', 'NSW', 'V'],
```

```
    'Q': ['NT', 'SA', 'NSW'],
    'NSW': ['Q', 'SA', 'V'],
    'V': ['SA', 'NSW'],
    'T': []
}

available_colors = ['Red', 'Green', 'Blue']
node_list = list(map_graph.keys())
color_assignment = {}

if map_coloring_csp(map_graph, available_colors, color_assignment, node_list):
    print("Valid map coloring assignment:")
    for territory, assigned_color in color_assignment.items():
        print(f"  {territory}: {assigned_color}")
else:
    print("No valid solution exists.")
```

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            return False
    return True

def map_coloring_csp(graph, colors, assignment, nodes, index=0):
    if index == len(nodes):
        return True

    current_node = nodes[index]
    for color in colors:
        if is_safe(current_node, color, assignment, graph):
            assignment[current_node] = color

            if map_coloring_csp(graph, colors, assignment, nodes, index + 1):
                return True
```

Output

```
Valid map coloring assignment:
  WA: Red
  NT: Green
  SA: Blue
  Q: Red
  NSW: Green
  V: Red
  T: Red
```